

Matthew Millward

07572 414907 | MMillward2012@gmail.com | [LinkedIn Profile](#) | [Portfolio Website](#)

EDUCATION

Durham University

Predicted Graduation Date: 2026

Bachelor of Science in Mathematics (Second-Year Student)

- First Class Honours (**85.6% average**).
- Relevant modules: Programming I, Linear Algebra I, Computational Tools for Engineers and Scientists, Data Science and Statistical Computing II, Mathematical Modelling II, Statistical Inference II.

Sir Graham Balfour

2016 - 2023

A-Levels: A*A*A*A*A in Maths, Chemistry, Physics, Further Maths and an EPQ.

GCSEs: 9/9/9/9/9/8/8/7 (incl. Maths, English, Science, Computer Science).

PROFESSIONAL WORK EXPERIENCE

Accounting Intern, Mercedes-Benz Stourbridge

October 2022

- Conducted analysis of financial data for the Midlands Mercedes branch using proprietary software. Utilised advanced Excel functions and internal reporting tools to streamline data processing and improve accuracy.
- Enhanced data handling and reporting skills in a fast-paced, detail-oriented environment.

PERSONAL/ACADEMIC PROJECTS

Handwritten Number Recognition (Python, Scikit-Learn)

- Designed and implemented an SVM to classify **1,797 handwritten-digit images** with an 80/20 train-test split.
- Achieved over a **96% classification accuracy** by fine-tuning hyperparameters (such as gamma).
- Evaluated performance using confusion matrices and displayed the misclassified digits to enhance model precision.

Atmospheric Carbon Prediction Model (MATLAB)

- Engineered a high-performance **advection-diffusion model** using NASA's MERRA2 data to predict black carbon dispersion during the 2020 Australian wildfires.
- Improved processing efficiency by 84%**, reducing data generation time from 80 seconds to 13 seconds, improving data handling techniques through MATLAB's vectorised operations and pre-allocations of arrays.

Gauge Cluster Analysis (Python, OpenCV)

- Developed a video analysis programme for real-time monitoring of 1080p gauge readings at 30fps using image pre-processing techniques (incl. colour segmentation, Gaussian blur).
- Increased data capture rate by 90%** through machine learning-driven dynamic centre detection and leveraging adaptive colour segmentation, **reducing inconsistencies by 40%**.
- Achieved near real-time processing with an analysis time of 2.5 seconds per frame, improving operational efficiency for continuous video analysis.

COVID-19 Spread Modelling and Analysis (Python)

- Engineered and calibrated **SIR** and **SIRS** models using UK and Brazil data, quantifying key **epidemiological parameters** such as growth rate ($\lambda = 0.294$) and doubling time ($\tau = 2.35$ days).
- Introduced advanced features such as **seasonal variation** and **immunity decay**, significantly improving model accuracy and predictive capabilities over traditional approaches.
- Delivered an **18-page technical report** detailing model development, assumptions, and results, effectively communicating complex findings to both technical and non-technical stakeholders.

DurHack 2023/24 (24-hour Hackathon)

- DurHack 2024: Led a team of 3 to develop a bar crawl route optimisation website using **Dijkstra's algorithm** and OpenStreetMaps API. Rapidly learned and implemented **Next.js** to transform complex geospatial data into an intuitive, interactive navigation tool that calculates the most efficient path between selected venues.
- DurHack 2023: Our team built a comprehensive e-commerce platform with a Tinder-like interface for product discovery. We developed a **full-stack** solution using HTML, CSS, and JavaScript for the front end, and Go for backend, integrating **50 product data entries** from THG, and creating a functional marketplace prototype within 24 hours.

TECHNICAL SKILLS

- Programming Languages:** Python, R, MATLAB, C, SQL, HTML, CSS.
- Libraries/Tools:** NumPy, Pandas, Scikit-learn, OpenCV, Flask, SQLite, Git, Matplotlib, Excel.
- Development tools:** Git (version control), Flask (web frameworks), SQL (databases).

CERTIFICATIONS

- EdX CS50x 2024 Introduction to Computer Science. **August - September 2024**
- Academy for PhD Training in Statistics (APTS) **November 2024 - Present**